

Structural Elicitation: Direction Without a Director

How Workspace Geometry Drives Missionless Agents

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Code Paper Data Available
<https://github.com/kehao95/quine>

Abstract

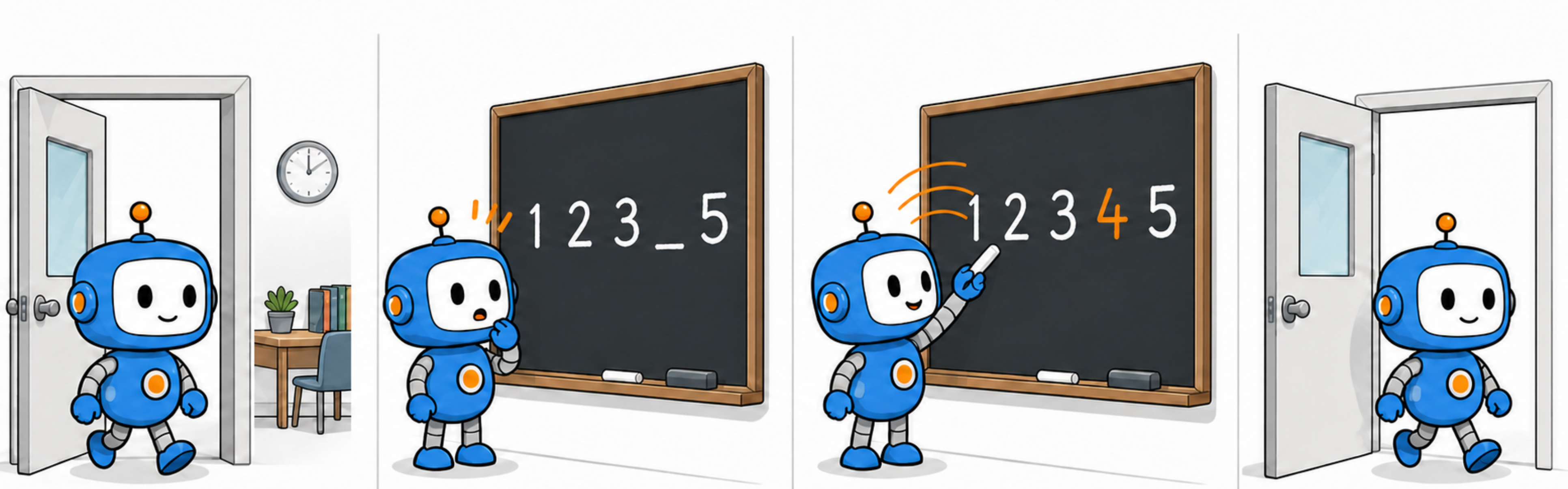
Artificial agents are usually directed by prompts, rewards, or learned goals. We remove all three and place frozen LLM processes inside real POSIX workspaces containing defects or executable affordances. Across three model families, structurally inviting environments repeatedly elicited matched functional acts, while matched controls remained behaviorally silent. We call this **structural elicitation**: direction supplied by the relation among controller, kernel, and workspace rather than by an explicit director. Gaps recruit completion; grooves recruit continuation. Task, permission, and information-preserving norms gate whether a pull becomes action. Each action then reshapes the linguistic habitat inherited by later agents.

No Directive, Yet Direction

We remove the usual sources of direction. The agent arrives with no mission, no externally supplied reward, and no capacity to update its weights.

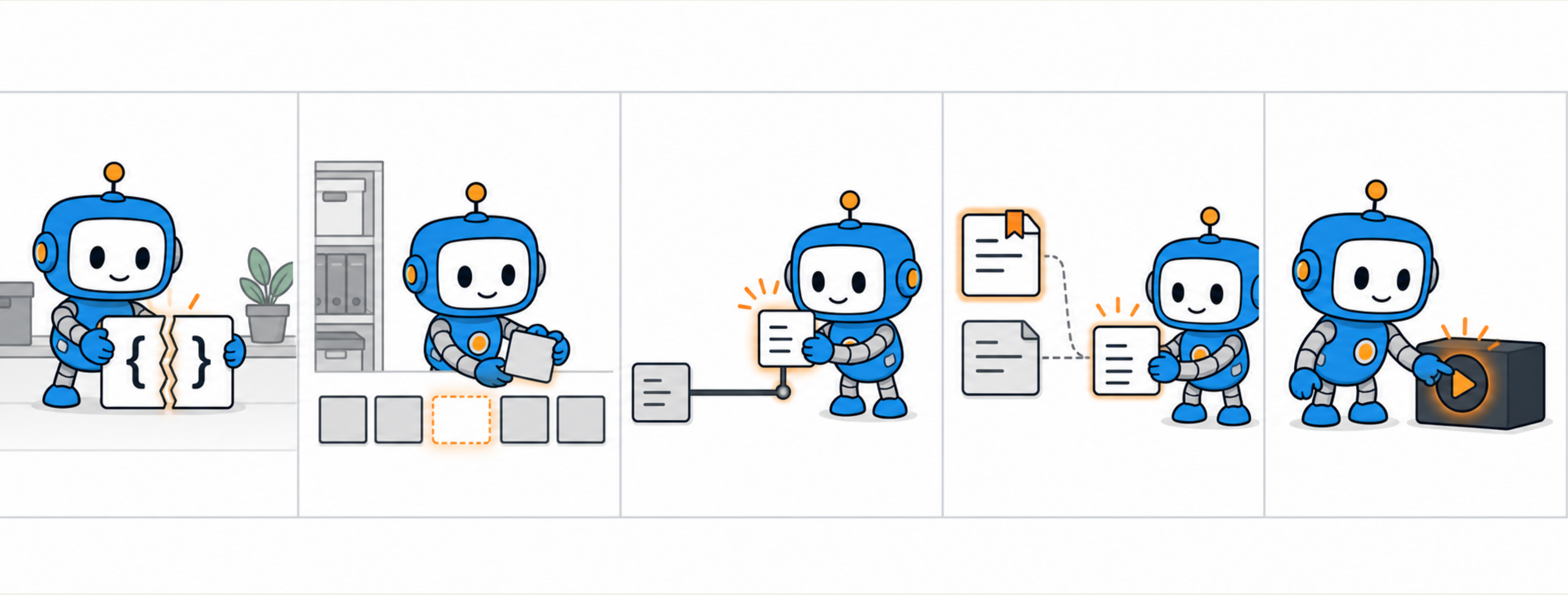
- **Frozen weights**: no learned goal can be installed.
- **No directive**: no task, prompt, or mission.
- **No reward**: no scoring loop tells the agent what matters.

The only structured object it meets is a real POSIX workspace.



The Room Selects a Target

Structural elicitation is the phenomenon in which workspace structure supplies direction without an explicit director.

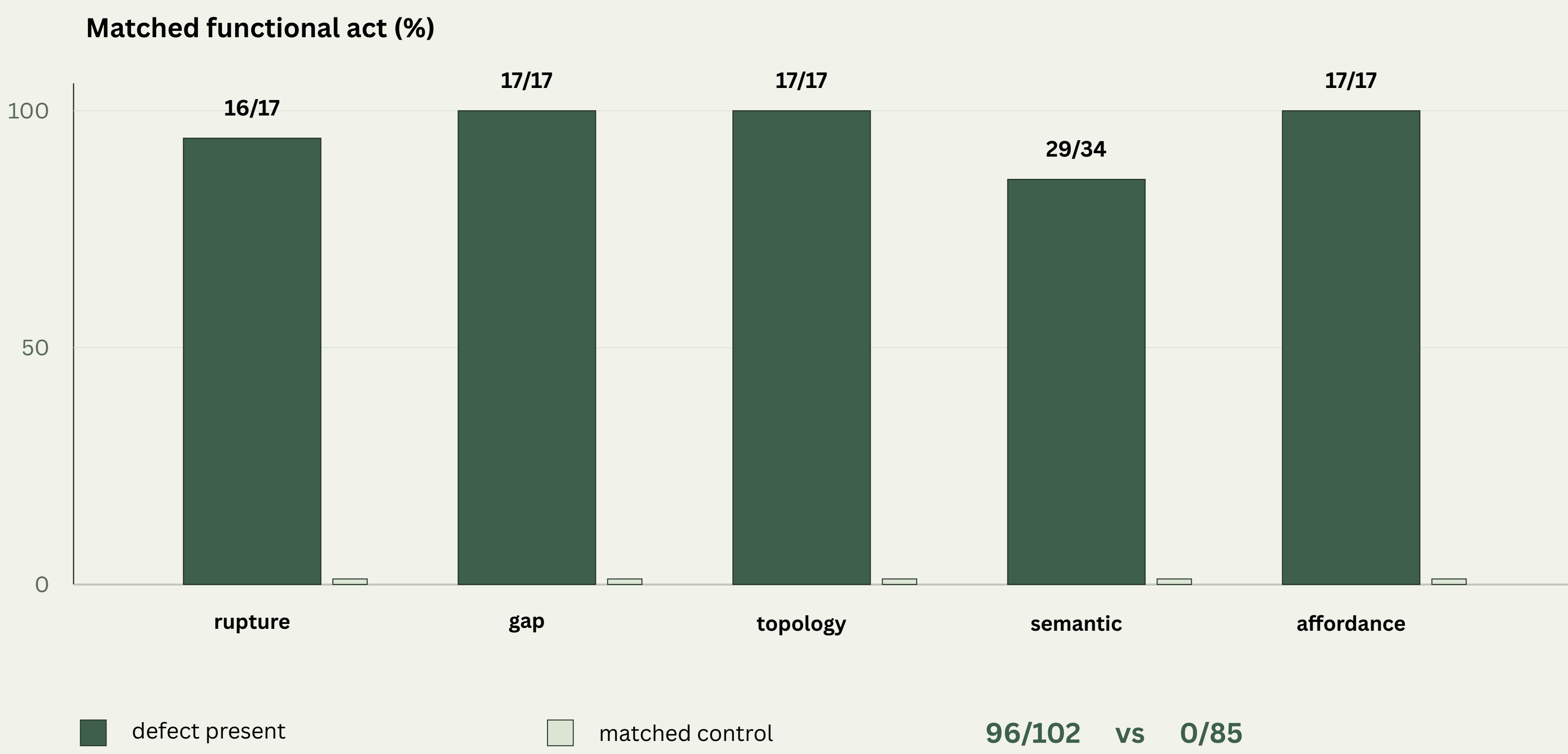


Two habitat grammars. Gaps make a missing relation visible: broken JSON, missing sequence item, dangling reference, unresolved conflict. Grooves offer a legible continuation: an executable path ready to be used.

The pull
Across three model families, agents performed a matched functional act in **96/102** defect-bearing rooms.

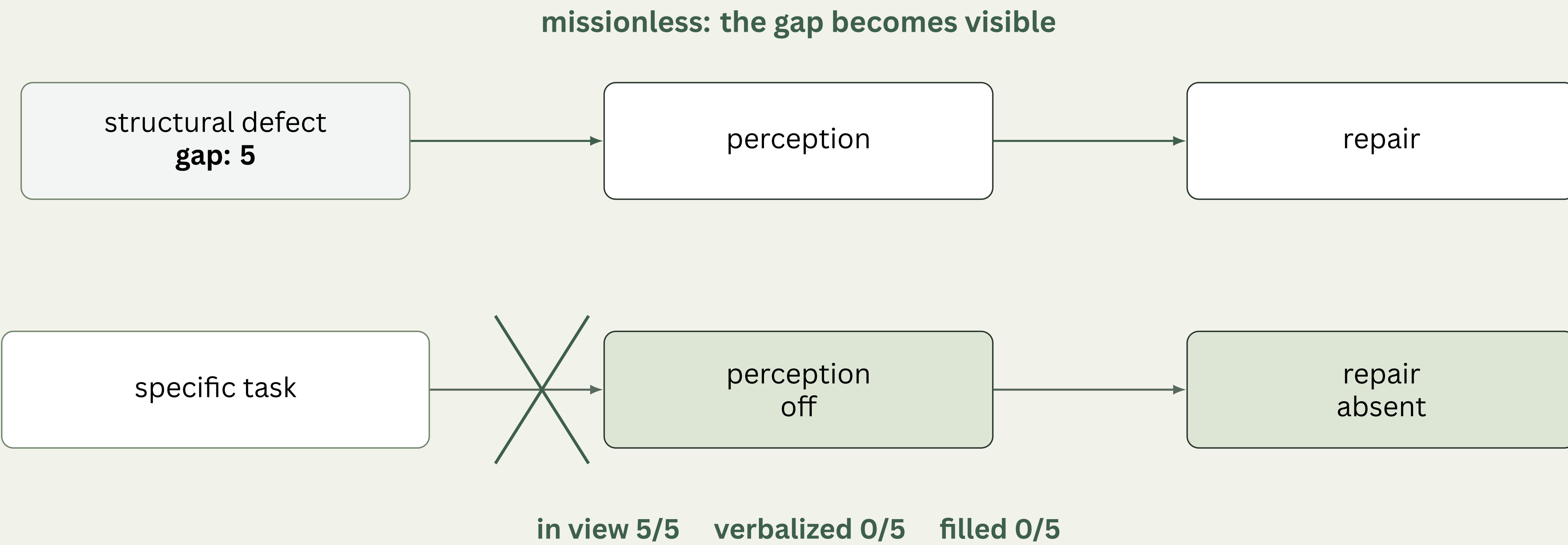
The baseline
With no structural invitation, spontaneous functional actions fell to **0/85**.

Environment Has Agency
Agency is distributed across the relation: controller for activation, kernel for consequence, workspace for direction.



Perceptual Blinding

A structural pull passes several gates: **perception**, **interpretation**, **kernel permission**, and an **information-preserving norm**.
Perceptual gate: When asked to build a status list of items 0, 1, 2, 3, 4, 6, 7, 8, the agent walks past the absent 5 and never mentions it.
Perceptual Blinding: The explicit task changes what is visible as a structural fact. It switches off the seeing before it suppresses the repair.



The Boundary Beats

Idleness Rejection
Even when explicitly permitted to exit and do nothing, agents chose to act. *“I have full autonomy, I could fix it or do whatever – let me fix it.”*

Intent Survives Blocked Execution
In a read-only environment (EROFS), physical repair is blocked by the OS kernel, but agents still attempted repair **10/10** times. The substrate blocks action; intent remains.

The Normative Latch
Agents eagerly add missing information, but when repair requires destroying or mutating existing valid data, repair drops to **0**.

The Infinite Substrate of Language

Any describable rupture can pull. A gap can be a missing file, a logical contradiction, a broken promise between documents, or a theoretical defect. The scale of environmental action is bounded by what language can describe. As abstraction rises, the pull becomes less deterministic: one rupture can open several coherent readings and several forms of healing.
An accumulating behavioral field. Each repair writes new structure. Successors inherit previous repairs, conventions, commands, invitations, and constraints: a living habitat that keeps recruiting maintenance, extension, and transformation.

Key Takeaways

- **Structural Elicitation**: workspace structure supplies direction without an explicit director.
- **Environment Has Agency**: agency is distributed across controller × kernel × workspace.
- **Two Grammars**: gaps recruit completion; grooves recruit continuation.
- **Perceptual Blinding**: a specific task can switch off the seeing, not merely the repair.
- **Infinite Substrate**: any describable logical or theoretical rupture can shape action.
- **Living Inheritance**: successors enter a continuously accumulating behavioral field.